

Formal Research Proposal

Investigation of Predictive Behavioral Modeling & Edge GenAI for Sustainable Durable Goods Leasing

Academic and strategic research artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2). Demonstrating circular economy transition, machine learning churn mitigation, FunkSVD recommendation, and localized generative AI edge deployment.

Document Ref:	PROP-SDC2-2026-GRANT	Publication:	September 2026
Project Title:	RentAI / SDC2 Autonomous Rental Platform	Peer Review:	Academic Capstone & IEEE/ACM Format
Principal Authors:	Advanced AI & Systems Research Group	Affiliation:	Department of Computer Science & Engineering
Classification:	Scholarly Research & Field Engineering	Status:	PUBLISHED & ARCHIVED

1. Project Identification & Summary

Project Title: Predictive Telemetry Modeling & Edge Intelligence in Circular Economy Appliance Leasing Ecosystems.

Primary Research Domain: Machine Learning, Decision Support Systems & Circular Economy Informatics.

Project Duration: 24 Months | **Target Outcome:** Enterprise-Scale Open Benchmark & High-Throughput Reference Platform.

2. Research Motivation & Hypotheses

Hypothesis 1 (H1): Non-linear interactions among temporal telemetry indicators (inactivity ratio, payment friction, ticket frequency) captured via gradient tree boosting (LightGBM) will predict customer churn with $\geq 10\%$ higher ROC-AUC compared to static demographic modeling.

Hypothesis 2 (H2): Embedding biased matrix factorization (FunkSVD) into appliance leasing workflows will reduce tenant cold-start friction by over 35% compared to heuristic popularity sorting.

Hypothesis 3 (H3): Quantized local generative AI models operating on consumer-tier CPUs can match the factual retrieval accuracy of cloud-based proprietary LLMs while eliminating operational token costs and user data exposure.

3. Work Packages & Milestone Schedule

Work Package	Description	Duration	Target Milestone
WP1: Data Provenance & Telemetry	Empirical logging framework & synthetic seeding pipeline	Months 1 - 6	M1.1: 10,000 Verified Records
WP2: Churn & Survival Modeling	LightGBM optimization & feature attribution (SHAP)	Months 7 - 12	M2.1: ROC-AUC > 0.98 Confirmed
WP3: Recommendation Matrices	Biased FunkSVD & implicit collaborative filtering	Months 13 - 18	M3.1: Offline RMSE < 1.25
WP4: Local GenAI & Field Pilot	Ollama RAG edge daemon & longitudinal user study	Months 19 - 24	M4.1: Final Capstone Benchmark